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Notes from Digital Image Processing

1. Before digital processing an image is fundamentally described as a 2D light intensity function $f(x, y)$. This function is a combination of illumination ($i(x, y)$) and reflectance ($r(x, y)$).
2. While computers commonly use the RGB model, the HSI model is more aligned with human perception. It separates the color-carrying information from the brightness, making it ideal for developing algorithms that mimic how we interpret color.
3. Image compression works differently for photos and drawings, using JPEG for complex photos and GIF for simple shapes and logos to save space.
4. Machines see images as giant grids of numbers, where each tiny square (a pixel) has a value representing its color or brightness.
5. Object recognition teaches computers to identify things like humans do, using machine learning to find patterns or deep learning to learn features on its own.

Types of healthcare reports and medical imaging reports

Healthcare Reports

- These are narrative or structured documents created by healthcare providers to document patient care.
 - History and Physical (H&P)
 - Consultation Report
 - Progress Notes (SOAP Notes)
 - Discharge Summary
 - Operative Report (Op Note)
 - Procedure Report
 - Pathology Report
 - Laboratory Report
 - Emergency Department (ED) Report

Medical Imaging Reports

- These are specialized reports created by radiologists or imaging specialists to interpret medical images.
- Common structure of an Imaging report
 1. **Patient & Exam Demographics:** Name, ID, date of birth, date of exam.
 2. **Clinical Indication/History:** Why the exam was ordered (e.g., Cough and fever, Follow-up lung nodule).
 3. **Comparison:** Prior relevant studies used for comparison (crucial for tracking changes).
 4. **Technique:** Brief description of how the exam was performed (e.g., CT chest performed without intravenous contrast).
 5. **Findings:** Detailed, organized description of what is seen, organ by organ. Uses precise anatomical and descriptive language.
 6. **Impression/Conclusion:** The most important section. Summarizes the key results, provides a diagnosis or differential diagnosis, and often recommends next steps.
- Some types of imaging reports
 - X-Ray Report
 - Computed Tomography (CT) Scan Report
 - Magnetic Resonance Imaging (MRI) Report
 - Ultrasound (US) Report
 - Nuclear Medicine & PET Scan Report
 - Mammography Report
 - Fluoroscopy Report